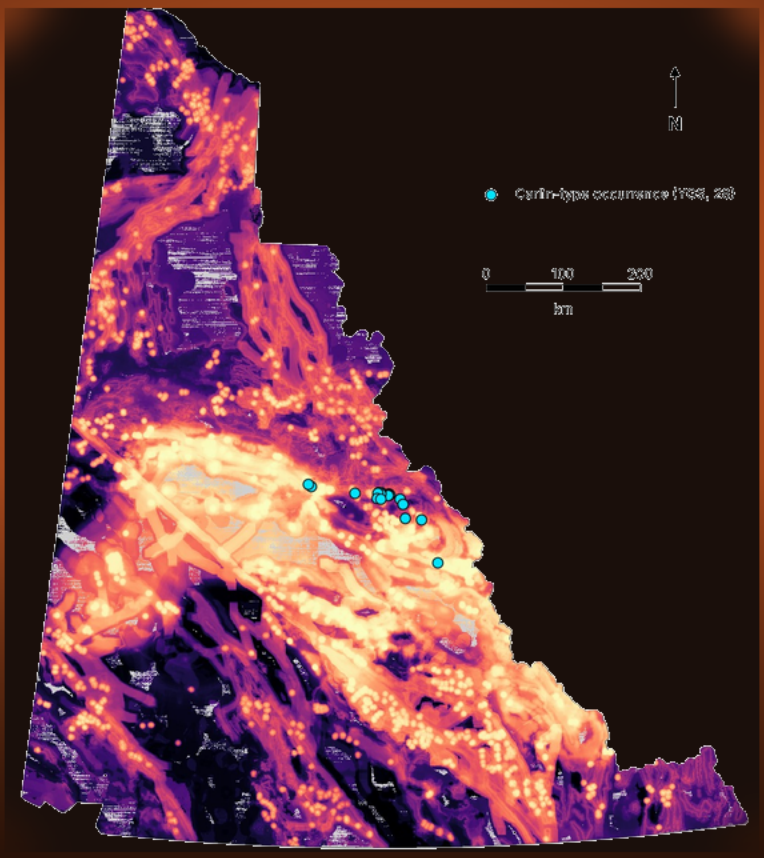
TEAM TERRA-X · USER GUIDE

INTELLIGENT PROSPECTING AGENT

From raw exploration data to a prospectivity map for new ground, even where no deposits are known.



19
EVIDENCE LAYERS

40
NEVADA DEPOSITS

0
LOCAL DEPOSITS

FIRST VERSION · CARLIN-TYPE GOLD

THE IDEA

WHAT THE AGENT DOES

The Intelligent Prospecting Agent is a proprietary, AI-native prospectivity engine developed in-house by Team Terra-X. It turns diverse exploration data into geological evidence layers, learns mineral-system signatures from known districts, and maps prospective ground in new regions.

1 ASK THE GEOLOGY

You give a study boundary and a descriptive model of the area. You approve every choice before anything is built.

2 BUILD EVIDENCE LAYERS

The agent builds predefined layers across Source, Energy, Pathway and Trap, like an evidence-layer workflow, but automated.

3 TRANSFER FROM NEVADA

A model learned on 40 Nevada Carlin deposits scores your ground. No local deposits are needed.

4 TEST IT HONESTLY

If you know some occurrences, it counts how many the map captures, and compares with chance and a simple map.

EXAMPLE · YUKON

Applied to the Yukon with 10 layers and no Yukon deposit used in training, the map placed 16 of 28 known Carlin-type occurrences in its top 10% and 21 of 28 in its top 20%. Chance would give about 3 and 6.

A simple map still scored a little higher, and most occurrences sit in one belt, so we call this an indicative result.

16 OF 28

IN THE TOP 10%

No local deposit used in training.

NEW GROUND HAS NO DEPOSITS TO LEARN FROM.
NEVADA HAS 84. THE AGENT CARRIES THEM ACROSS.

Every layer is visible, every proxy is labelled, and every result is tested against chance.

IN SIX STEPS

FROM DATA TO MAP

You need a study boundary, a short geology description, and your map data. Known occurrences are optional and are used only to check the result.

STEP 1 · INSTALL

Add carlin-prospecting-engine.skill in Claude's Skills settings, or run the Python scripts from the repository.

STEP 2 · DEFINE THE AREA

Give the boundary of ground where Carlin-type deposits can form, and a paper or notes on its geology and the age of the ore.

STEP 3 · ADD YOUR DATA

Geology map, faults, and geochemistry if you have it. The agent inspects the files and reports problems before building.

STEP 4 · APPROVE THE LAYERS

For each layer, accept, drop or change it. Source and Energy are built from your own intrusion ages.

STEP 5 · RUN

The agent scores your ground against the Nevada signature and writes a prospectivity map you can open in QGIS.

STEP 6 · CHECK

Read the summary: layers used, proxies, areas outside Nevada's range, and the capture of known occurrences.

WHAT YOU GET BACK

A prospectivity map (GeoTIFF) with a percentile for every cell, one image per evidence layer, a plain-language summary with the verdict, and, when you supply known occurrences, a table of how many the map captures.

IT MAPS SIMILARITY, NOT DEPOSITS.

It is not a deposit probability and makes no claim about tonnage or economics. Check every result against a simple map, and treat one district as a first test, not proof.

Repository: see the README for the folder layout, install steps and limitations.